Universidade de São Paulo
Escola Politécnica - Engenharia de Computação e Sistemas Digitais

Clustering

Prof. Artur Jordão

Introduction

Clustering

- Clustering is a problem of learning to assign a label to examples by leveraging an **unlabeled** dataset

- It refers to a very broad set of techniques for finding **subgroups**, or clusters, in a dataset

- Clustering is often used to analyze very large data sets
  - For example, in the context of social network analysis, clustering algorithms attempt to identify natural communities within large groups of people

Introduction

Clustering

- When we cluster the samples of a dataset, we seek to partition them into distinct groups
  - The observations within each group are quite similar to each other
  - The observations in different groups are quite different from each other

- To formalize this idea, we must define what it means for two or more observations to be similar or different
  - Distance metrics

Popular Algorithms

Clustering

- k-means
- Hierarchical Clustering
- DBSCAN
- Spectral Clustering

K-Means

Problem Definition

K-Means

- Let $c_1, \dots c_k$ denote sets containing the samples in each cluster
    - We will use k as a cluster label
    - If the a sample x_i is in the kth cluster, then the **label of $\boldsymbol{x_i}$ is $\boldsymbol{k}$**

- These sets satisfy two properties
    - $c_1 \cup c_2 \dots \cup c_k = \{x_1, \dots x_n\}$. This mean each sample $x \in X^n$ belongs to at least one of the k clusters
    - $c_k \cap c_{k'} = \emptyset$ for all $k \neq k'$. This means the clusters are non overlapping: no samples belong to more than one cluster

- Note that, the clustering process **does not employ $\boldsymbol{Y}$**

Problem Definition

K-Means

- The idea behind k-means clustering is that a good clustering is one for which the within-cluster variation is as small as possible

- Formally, $min_{c_1,\ldots.c_k}\left\{\sum_{j=1}^{k}\mathcal{M}(c_j)\right\}$
  - $\mathcal{M}(.)$ measures **the within-cluster variability**
  - The most common choice involves squared Euclidean distance as $\mathcal{M}(.)$
  - $\mathcal{M}(c_k) = \frac{1}{|c_k|}\sum_{x_i \in c_k}||x_i - u_k||^2$, where μ_k is the average of cluster c_k

- The previous formulas say that we want to partition the samples into k clusters such that the total within-cluster variation, over all k clusters, is as small as possible

General Idea

K-Means

- Choose the number of clusters k

- Randomly insert k samples (**centroids**) in the feature space
  - $m_1, \dots m_k$
  - **Initial seeds**

- Compute the distance from each example $x_i \in X$ to each centroid m_k using some distance metric (e.g., Euclidean distance)

- Assign each sample to the cluster whose centroid is closest
  - We assign to each sample the identifier k of its corresponding centroid as its **label**

General Idea

K-Means

- For each cluster, calculate the average feature vector of the samples within it
  - $m_k = \frac{1}{|c_k|} \sum_{x_i \in c_k} x_i$
  - **These average feature vectors become the new locations of the centroids**
  - Note that, after the first step, $m_k = \mu_k$

- Recompute the distance from each sample to each centroid and modify the assignment
  - Note that, at this step, the centroid locations may change

Hands-on

- Repeat the previous steps until:
  - The assignments don't change after the centroid locations
  - Achieve a pre-defined number of steps
  - The decrease in $\sum_{j=1}^{k} \mathcal{M}(c_j)$ is below a pre-defined threshold

Algorithm

K-Means

- Cluster initialization
  - Randomly create k samples (centroids)
  - Randomly assign a number, from 1 to k, to each of the observations (here, swap 2.1 to 2.2)
  - Randomly select k samples from X

K-Mean Clustering

1. Initialize k centroids (insert k centroids in the feature space)
2. Iterate until the cluster assignments stop changing

   2.1 Assign each sample x_i to the closest centroid

   2.2 Update each centroid as the mean of the points assigned to it $\lhd \frac{1}{|c_k|} \sum_{x_i \in c_k} x_i$

Practical Issues

K-Means

- The k-mean algorithm needs to compute all distances at every iteration
- The initial position of centroids influence the final positions
  - Different runs can result in different clusters
- The algorithm is sensitive to initial seeds and outliers

Hands-on

The Role of Parameter k

K-Means

- An important question is *how many clusters does your dataset have?*

- For any chosen k, clustering always finds k centers, regardless of whether they represent meaningful groups

- If k is too small
  - Any new sample falls into the same group
  - No useful clustering emerges

- If k is too large
  - Every sample becomes a cluster

Hierarchical Clustering

Introduction

Hierarchical Clustering

- One potential disadvantage of K-means clustering is that it requires us to pre-specify the number of clusters k

- Hierarchical clustering is an alternative approach and does not require committing to a specific value of K

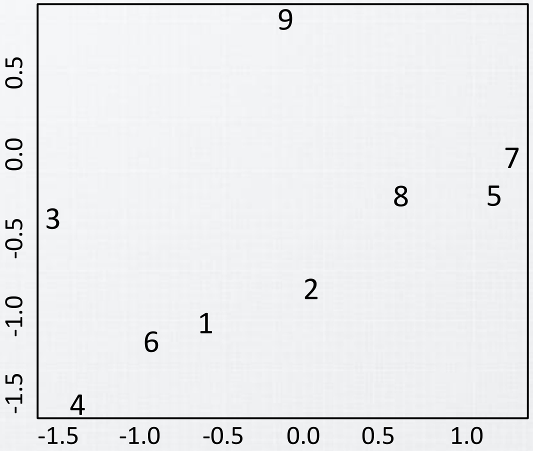
- It also offers an advantage over K-means clustering by producing a tree-based representation of the observations
  - Dendrogram

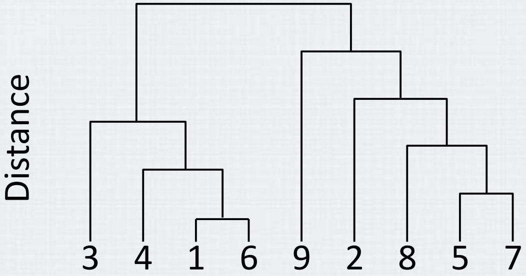
Dendrogram

Hierarchical Clustering

- Each **leaf** of the dendrogram represents one sample

- As the tree moves upward, some **leaves** fuse into branches, corresponding to samples that are similar to each other



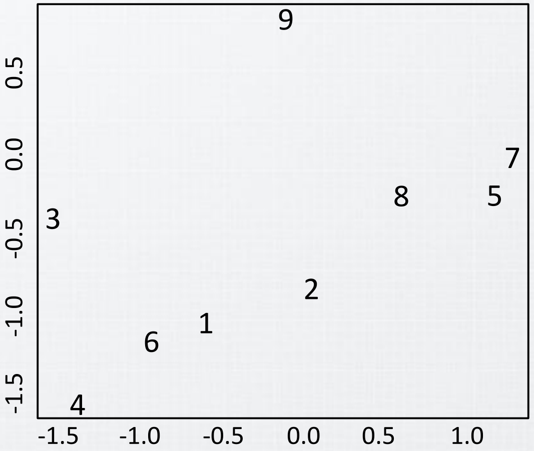


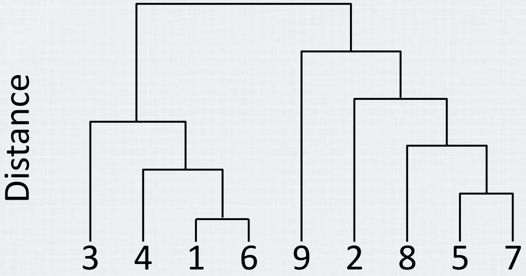
Dendrogram

Hierarchical Clustering

- The earlier (lower in the tree) fusions occur, the more similar the groups of observations are to each other
    - Samples that fuse at the very bottom of the tree are quite similar to each other

- Samples that fuse later (near the top of the tree) can be quite different
    - Samples that fuse close to the top of the tree will tend to be quite different

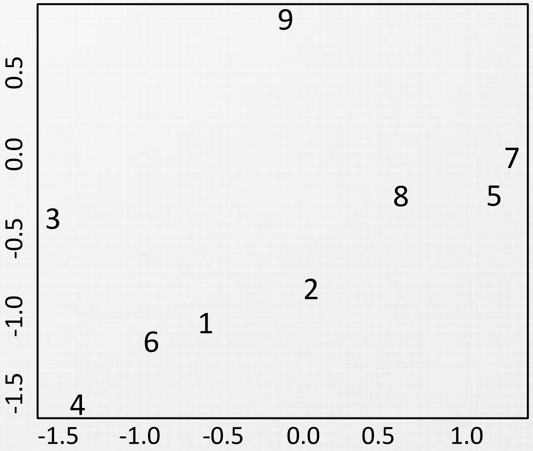


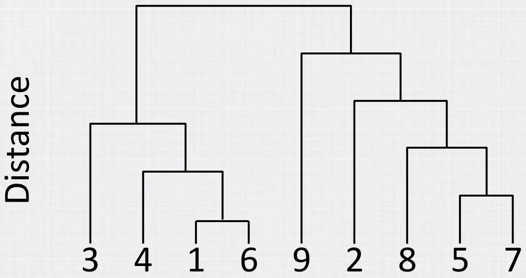


Dendrogram

Hierarchical Clustering

- The height of the fusion (vertical axis) indicates the dissimilarity between the two samples (or clusters)





Algorithm

Hierarchical Clustering

Hierarchical Clustering (X)

Treat each sample as its own cluster

For $i = n, n-1, \ldots 2$

1. Examine all pairwise inter-cluster dissimilarities among the i clusters
2. Identify the pair of clusters that are least dissimilar (most similar)
3. Fuse these two clusters ⊲ The dissimilarity between these two clusters indicates the height in the dendrogram at which this fusion occurs
4. Compute the new pairwise inter-cluster dissimilarities among the $i - 1$ remaining clusters

Overview

Hierarchical Clustering

- How do we define the dissimilarity between two clusters if one or both of the clusters contains multiple samples?
  - The concept of dissimilarity between a pair of samples needs to be extended to a pair of groups of observations
- Linkage
  - It defines the dissimilarity between two groups of samples

Linkage

Hierarchical Clustering

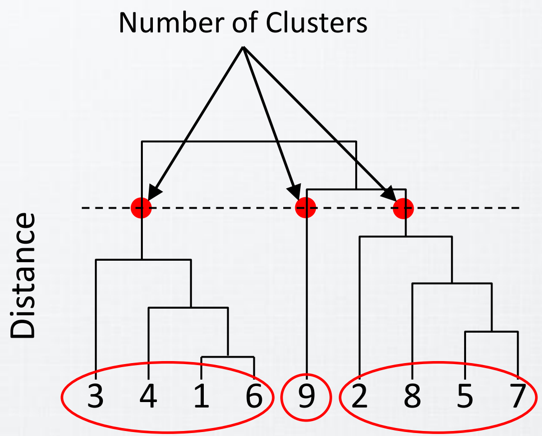
Linkage	Summary	Formula
Complete (or maximum)	Maximal intercluster dissimilarity. Compute all pairwise dissimilarities between the samples in cluster A and the samples in cluster B, and record the largest of these dissimilarities	$d(A,B) = \max_{a \in A, b \in B}(a,b)$
Single (or minimum)	Minimal intercluster dissimilarity. Compute all pairwise dissimilarities between the samples in cluster A and the samples in cluster B, and record the smallest of these dissimilarities	$d(A,B) = \min_{a \in A, b \in B}(a,b)$
Average	Mean intercluster dissimilarity. Compute all pairwise dissimilarities between the samples in cluster A and the samples in cluster B, and record the average of these dissimilarities	$d(A,B) = \frac{1}{\lvert A \rvert \lvert B \rvert} \sum_{a \in A} \sum_{b \in B} d(a,b)$
Centroid	Dissimilarity between the centroid (μ) for cluster A (a mean sample of length $1 \times m$) and the centroid for cluster B	$d(\mu_A, \mu_B)$

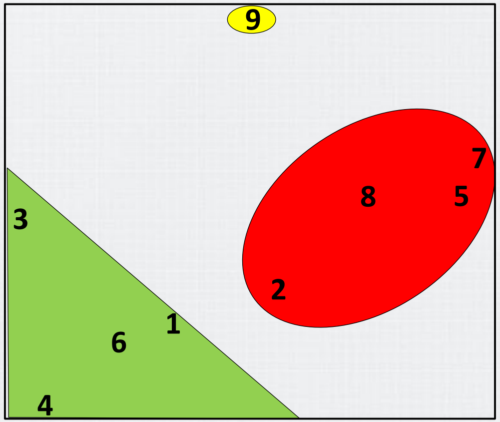
Hands-on

Distance Cut

Hierarchical Clustering

- The distance cut results in different clusters





Hands-on

Practical Issues

Hierarchical Clustering

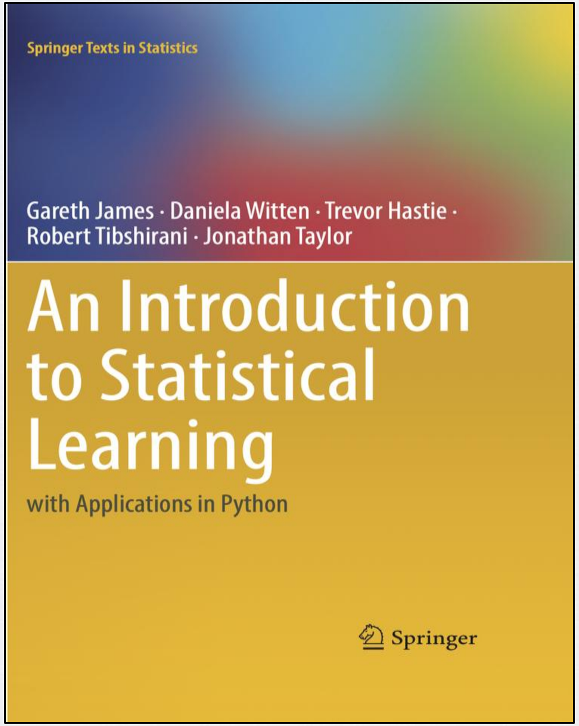
- Which dissimilarity measure is appropriate?

- Which linkage criterion is appropriate?

Bibliography

Bibliography

- An Introduction to Statistical Learning
    - Chapter 10
        - 10.3.1 K-Means Clustering
        - 10.3.2 Hierarchical Clustering



Bibliography

- INTRODUCTION TO Machine Learning
  - Chapter 7
    - 7.3 k-Means Clustering
    - 7.8 Hierarchical Clustering